



KILLARA HIGH SCHOOL

**YEAR 12 TERM 1 ASSESSMENT
2007**

EXTENSION 1 MATHEMATICS

Time Allowed: 2 hours

Total marks: 84

- * All questions are of equal value.
- * Start each question in a new booklet.
- * Write on one side of the paper only.
- * Show all necessary working.

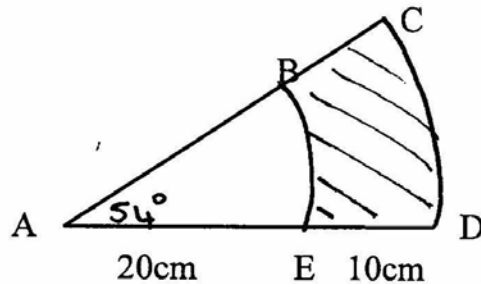
Question 1 (12 marks)

Marks

- (a) (i) Convert 54° to radians (leave answer in terms of π).

1

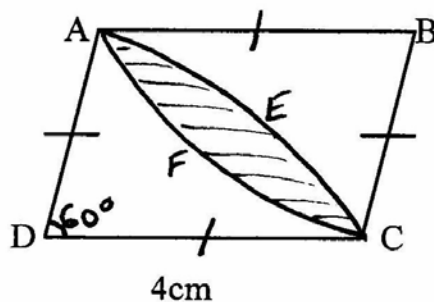
(ii)



CD and BE are arcs of circles with centre A.
AE = 20cm, ED = 10cm.
Find the perimeter of the shaded region BCDE.

2

(b)



ABCD is a rhombus of side length 4cm. AEC and AFC are arcs of circles with centres B and D respectively.

- (i) Show that the area of the rhombus ABCD is $8\sqrt{3}$ cm².

1

- (ii) Find the area of the shaded region.

2

- (c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{4x}$

1

- (d) Differentiate (i) $x \tan x$

1

- (ii) $\sin^3 x$

2

- (e) By expressing $\cot x$ in terms of $\sin x$ and $\cos x$, or otherwise, show that the derivative of $\cot x$ is $-\operatorname{cosec}^2 x$.

2

Continued over page

Question 2 (12 marks)**Marks**

(a) Find primitive functions of

(i) $\sin(4x-3)$

1

(ii) $\tan^2 x$

2

(iii) $\cos^2 x$

2(b) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos 2x \, dx$ **2**(c) A particle is moving in Simple Harmonic Motion. Its displacement x metres at time t seconds is given by $x = 2 \cos\left(3t + \frac{\pi}{5}\right)$.

(i) What is the period and amplitude of the motion?

2(ii) Show that $\ddot{x} = -9x$ **2**

(iii) Find the maximum speed of the particle.

1

Question 3. (12 marks)**Marks**

- (a) The acceleration, $a \text{ m/s}^2$, of a particle moving in a straight line is given by $a = -e^{-2x}$ where x metres is the displacement from the origin at time t seconds. Initially the particle is at the origin with velocity 1 m/s . Find the displacement as a function of time.

5

- (b) A projectile is fired with initial speed $V \text{ m/s}$ at an angle α to the horizontal.

- (i) Given that the projectile's position is given by $x = Vt \cos \alpha$, $y = Vt \sin \alpha - \frac{1}{2}gt^2$ (you do not need to prove this), show that the equation of the path of the projectile is

3

$$y = x \tan \alpha - \frac{gx^2}{2V^2 \cos^2 \alpha}$$

- (ii) Show that the range is given by $x = \frac{V^2 \sin 2\alpha}{g}$

2

- (iii) Find the maximum range.

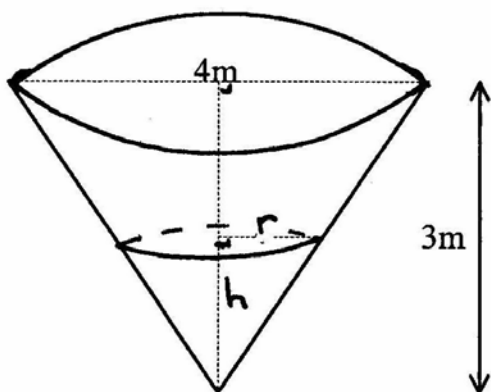
1

- (iv) Explain the effect on the maximum range if the initial speed is doubled, all other factors remaining the same.

1

Question 4 (12 marks)**Marks**

(a)



Water is pouring into a large storage tank which is conical in shape. The diameter of the tank at its top is 4m and its height is 3m. When the height of water is h m the radius of the tank is r m.

- (i) Use similar triangles to show that $r = \frac{2h}{3}$ 1
- (ii) Hence show that the volume, V , of water in the tank can be expressed as $V = \frac{4\pi h^3}{27}$ 1
- (iii) If water is flowing into the tank at a constant rate of $0.008 \text{ m}^3/\text{s}$, show that the height of water in the tank is increasing at the rate $\frac{dh}{dt} = \frac{0.018}{\pi h^2}$ 2
- (iv) At what rate is the water level rising when the tank is half full? 1
- (b) The rate at which a substance cools is proportional to the difference between its temperature and the temperature of the surrounds, ie $\frac{dT}{dt} = k(T - S)$ where T is the temperature of the substance, S is the temperature of the surrounds, t is the time in minutes and k is a constant.
- (i) Show that $T = S + Ae^{kt}$, where A is a constant, satisfies the equation. 1
- (ii) On a day when the temperature is 28°C , a substance cools from 100°C to 80°C in 15 minutes. What will be the temperature of the substance after a further 15 minutes? 4
- (iii) Sketch a graph of temperature T at time t , showing its main features. 2

Continued over page ...

Question 5. (12 marks)**Marks**

- (a) In how many ways can 6 people be arranged
- (i) in a straight line? **1**
 - (ii) in a circle? **1**
- (b) There are 7 swimmers in a race. In how many ways can 1st, 2nd and 3rd place be filled? **1**
- (c) Simplify $\frac{{}^nP_5}{{}^nC_3}$ **3**
- (d) How many arrangements of the letters of the word KILLARA are possible? **2**
- (e) A committee of 4 people is to be chosen from 9 members of a social club. Bob and Jenny are 2 of the 9 members. How many different committees is it possible to form
- (i) if there are no restrictions **1**
 - (ii) containing both Bob and Jenny **1**
 - (iii) containing neither Bob nor Jenny **1**
 - (iv) containing at least one of Bob and Jenny **1**

Continued over page ...

Question 6. (12 marks)**Marks**

- (a) Find the quotient and remainder when $P(x) = x^4 - 3x^3 + 7x^2 - 5x - 2$ is divided by $A(x) = x^2 + 1$. **3**
- (b) The polynomial $P(x) = ax^3 + bx^2 - 8x - 28$ has a zero at -2 and leaves a remainder of -27 when divided by $x - 1$. Find the values of a and b . **3**
- (c) The polynomial equation $px^3 + qx^2 + rx + s = 0$ has roots which are in the ratio $2 : 3 : 4$. Show that $4qr = 39ps$. **4**
- (d) Draw a neat sketch, showing the nature of the curve at the x -intercepts, of $y = x^3(x - 2)^2$. **2**

Question 7. (12 marks)**Marks**

(a) Differentiate

(i) $e^{\cos x}$

1

(ii) $\log_e \sin x$

1(b) (i) Differentiate $\ln(\ln x)$ **1**

(ii) Hence evaluate $\int_e^{e^2} \frac{dx}{x \ln x}$

3(c) Find the constant K such that $y = e^x \cos x$ satisfies

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + ky = 0$$

3(d) Use the substitution $u = \tan x$ to find $\int \frac{\sin^3 x}{\cos^5 x} dx$ **3****END OF PAPER**

$$1) 54^\circ = \frac{54\pi}{180} = \frac{3\pi}{10} \quad \checkmark$$

$$) BE = 20 \times \frac{3\pi}{10}, \quad CD = 30 \times \frac{3\pi}{10} \quad \checkmark$$

$$) \text{ perimeter} = 6\pi + 9\pi + 30 \text{ cm} \\ (= 67.12 \text{ cm.}) \quad \checkmark$$

$$) (ii) A = (\frac{1}{2} \times 4 \times 4 \times \sin 60^\circ) \times 2 \\ = 8\sqrt{3} \text{ cm}^2 \quad \checkmark$$

$$) \text{ Sector AEC} = \frac{1}{2} \times 4^2 \times \frac{\pi}{3} \\ = \frac{8\pi}{3} \text{ cm}^2 \quad \checkmark$$

$$\text{Segment AEC} = \frac{8\pi}{3} - \frac{1}{2} \times 4^2 \times \sin 60^\circ \\ = \frac{8\pi}{3} - 4\sqrt{3} \quad \checkmark$$

$$\text{Shaded area} = 2 \times \text{Segment AEC} \\ = 16\pi - 8\sqrt{3} \text{ cm}^2 \\ (= 2.899) \quad \checkmark$$

$$) = \frac{3}{4} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \\ = \frac{3}{4} \quad \checkmark$$

$$) i) x \sec^2 x + \tan x \quad \checkmark$$

$$ii) 3 \sin^2 x \cos x \quad \checkmark \checkmark$$

$$) \frac{d}{dx} \frac{\cos x}{\sin x} = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} \\ = \frac{-1}{\sin^2 x} \\ = -\operatorname{cosec}^2 x \quad \checkmark$$

$$Q.2 a) i) -\frac{1}{4} \cos(4x-3) + C \quad \checkmark$$

$$ii) \int \tan^2 x \cdot dx = \int (\sec^2 x - 1) dx \\ = \tan x - x + C \quad \checkmark$$

$$iii) \int \cos^2 x \cdot dx = \int (\frac{1}{2} + \frac{1}{2} \cos 2x) dx \\ = \frac{1}{2} x + \frac{1}{4} \sin 2x + C \quad \checkmark$$

$$b) \frac{1}{2} \left[\sin \frac{3\pi}{2} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\ = \frac{1}{2} \left(\sin \frac{3\pi}{2} - \sin \frac{\pi}{2} \right) \\ = \frac{1}{2} (\frac{\sqrt{3}}{2} - 1) \quad \checkmark$$

$$c) i) T = \frac{2\pi}{\omega}, \quad \omega = 2 \quad \checkmark$$

$$ii) \dot{x} = -6 \sin(3t + \frac{\pi}{5}) \\ \ddot{x} = -18 \cos(3t + \frac{\pi}{5}) \\ = -9 (2 \cos(3t + \frac{\pi}{5})) \quad \checkmark$$

$$= -9x \\ \text{(iii) Max speed} = 4 \quad \checkmark$$

$$3a) a = -e^{-2x}$$

$$\frac{1}{2} v^2 = \frac{1}{2} e^{-2x} + C \quad \checkmark$$

$$v=1 \text{ when } x=0 \Rightarrow \frac{1}{2} = \frac{1}{2} + C \quad \checkmark$$

$$\frac{1}{2} v^2 = \frac{1}{2} e^{-2x}$$

$$v^2 = e^{-2x} \\ v = \pm e^{-x} \quad \checkmark$$

Since initially v is moving to the right $\therefore v$ is $\text{m/s} > 0$, take $v = e^{-x}$

$$\therefore \frac{dv}{dx} = e^{-x} \\ t = e^x + C \quad \checkmark$$

$$\text{When } t=0, x=0 \Rightarrow 0 = 1 + C$$

$$\therefore t = e^x - 1 \\ e^x = t + 1 \\ x = \ln(t+1) \quad \checkmark$$

$$) i) \text{ Subst. } t = \frac{x}{v_{\text{end}}} \text{ into } \checkmark$$

$$y = vt \sin \alpha - \frac{1}{2} g t^2 \\ = v_{\text{end}} \cdot \frac{x}{v_{\text{end}}} - \frac{g}{2} \times \frac{x^2}{v_{\text{end}}^2} \\ = x \tan \alpha - \frac{g x^2}{2 v_{\text{end}}^2 \cos^2 \alpha} \quad \checkmark$$

$$) \text{ Find } x \text{ when } y=0 \\ x \left(\tan \alpha - \frac{g x}{2 v_{\text{end}}^2 \cos^2 \alpha} \right) = 0 \\ x=0 \text{ or } \checkmark$$

$$\frac{gx}{2v^2 \cos^2 \alpha} = \tan \alpha$$

$$x = \frac{2v^2 \cos^2 \alpha \tan \alpha}{2v^2 \cos^2 \alpha} \\ = \frac{2 \tan \alpha}{2} \\ = \tan \alpha \quad \checkmark$$

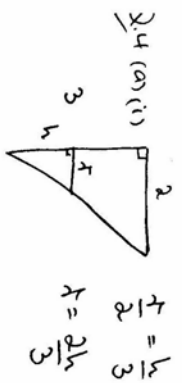
$$(iii) \text{ Max. range} = \frac{v^2}{g} \quad \checkmark$$

$$(iv) \text{ If } v \text{ is doubled} \\ \text{max. range} = \frac{(2v)^2}{g} \\ = 4 \frac{v^2}{g} \quad \checkmark$$

ie. Max. range is quadrupled.

$$\text{or (iii) Put } y=0 \Rightarrow \\ vt \sin \alpha - \frac{1}{2} g t^2 = 0 \\ t (v \sin \alpha - \frac{1}{2} g t) = 0 \\ t=0, t = \frac{2v \sin \alpha}{g} \quad \checkmark$$

$$\text{Subst. into } x = vt \cos \alpha \\ = v \cos \alpha \times \frac{2v \sin \alpha}{g} \\ = \frac{2v^2 \sin \alpha \cos \alpha}{g} \\ = \frac{v^2 \sin 2\alpha}{g} \quad \checkmark$$



$$1) V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{2h}{3}\right)^2 h$$

$$= \frac{4\pi h^3}{27}$$

$$i) \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$0.008 = \frac{4\pi h^2}{27} \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.008 \times \frac{27}{4\pi h^2}$$

$$= \frac{0.018}{\pi h^2}$$

ii) When $h = 1.5$, $\frac{dh}{dt} = \frac{0.018}{\pi \times 1.5^2}$
 $= 0.002545$

$$i) \frac{dT}{dt} = k(A - T)$$

$$= k(T - S)$$

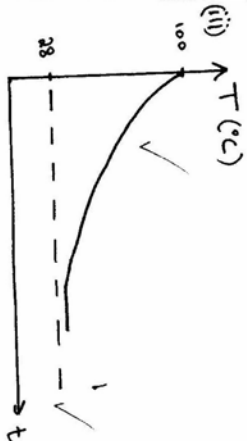
ii) $S = 28$, when $t = 0$ $T = 100$
 $\Rightarrow 100 = 28 + Ae^0$

$$\Rightarrow A = 72$$

$$T = 28 + 72e^{-kt}$$

When $t = 15$ $T = 80 \Rightarrow$
 $80 = 28 + 72e^{-15k}$
 $52 = 72e^{-15k}$
 $k = \frac{1}{15} \ln \frac{52}{72}$
 $= -0.0216 \dots$

When $t = 30$, $30 \times -0.0216 \dots$
 $T = 28 + 72e^{-0.648}$
 $= 66^\circ \text{C (to nearest degree)}$



1.5 (a) (i) 6! (iii) 5!

i) $7P_3 = 210$

i) ${}^nP_5 = \frac{n!}{(n-5)!} \div \frac{n!}{(n-3)! 3!}$
 $= \frac{n!}{(n-5)!} \times \frac{(n-3)! 3!}{n!}$
 $= \frac{3! (n-3)(n-4)}{(n-5)!}$
 $= 6(n^2 - 7n + 12)$

i) $\frac{7!}{2! 2!} = 1260$

i) ${}^9C_4 = 126$

(ii) ${}^7C_2 = 21$

(iii) ${}^7C_4 = 35$

(iv) ${}^9C_4 - {}^7C_4 = 91$ ((i) - (iii))

26

$$\frac{x^2 + 1}{x^4 - 3x^3 + 7x^2 - 5x - 2}$$

$$\frac{x^2 - 3x + 6}{x^4 - 3x^3 + 7x^2 - 5x - 2}$$

$$\frac{-3x^3 + 6x^2 - 5x + 6}{x^4 - 3x^3 + 7x^2 - 5x - 2}$$

$$\frac{6x^2 - 2x - 2}{x^4 - 3x^3 + 7x^2 - 5x - 2}$$

$$\frac{6x^2 - 2x - 2}{6x^2 - 2x - 2}$$

$$\frac{-2x - 8}{6x^2 - 2x - 2}$$

$Q(x) = x^2 - 3x + 5$
 $R(x) = -2x - 7$

(b) $P(-2) = 0 \Rightarrow -8a + 4b + 16 - 28 = 0$
 $\Rightarrow 2a - b = -3 \dots (1)$

$P(1) = -27 \Rightarrow a + b - 8 - 28 = -27$
 $\Rightarrow a + b = 9 \dots (2)$

From (1) + (2) $a = 2, b = 7$

c) Let roots be $2a, 3a, 4a$.

$2a + 3a + 4a = -\frac{q}{p}$ i.e. $9a = -\frac{q}{p}$

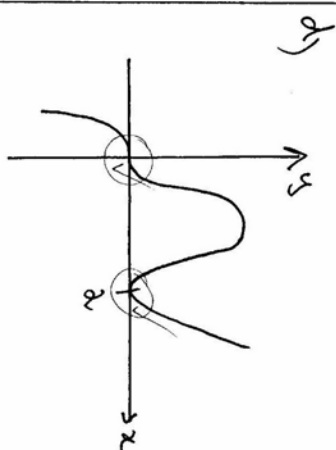
$6a^2 + 12a^2 + 8a^2 = \frac{r}{p}$ i.e. $26a^2 = \frac{r}{p}$

$24a^2 = -\frac{D}{p}$

$4q^2 = 4x - 9p^2 \times 26p^2$
 $= -936p^2 x^3$

$39ps = 39p^2 x - 24p^2 x^3$
 $= -936p^2 x^3$

$\therefore 4q^2 = 39ps$



$$b) (a) \text{ i) } -\sin x e^{\cos x} \quad \checkmark$$

$$(ii) \frac{\cos x}{\sin x} (= \cot x) \quad \checkmark$$

$$c) \text{ i) } \frac{1}{x \ln x} \quad \checkmark$$

$$\begin{aligned} \text{ii) } \int_e^{e^2} \frac{dx}{x \ln x} &= \left[\ln(\ln x) \right]_e^{e^2} \\ &= \ln(\ln e^2) - \ln(\ln e) \\ &= \ln 2 - \ln 1 \\ &= \ln 2. \quad \checkmark (= 0.69 \dots) \end{aligned}$$

$$\begin{aligned} \text{1) } y &= e^x \cos x \\ y' &= -e^x \sin x + e^x \cos x \quad \checkmark \\ y'' &= -e^x \cos x - e^x \sin x \\ &\quad - e^x \sin x + e^x \cos x \\ &= -2e^x \sin x \quad \checkmark \end{aligned}$$

$$y'' - 2y' + ky = 0$$

$$-2e^x \sin x + 2e^x \sin x - 2e^x \cos x + k(e^x \cos x) = 0$$

$$ke^x \cos x - 2e^x \cos x = 0 \quad \checkmark$$

$$k = 2. \quad \checkmark$$

$$d) \text{ If } u = \tan x$$

$$\begin{aligned} du &= \sec^2 x \, dx \\ &= \frac{dx}{\cos^2 x} \quad \checkmark \end{aligned}$$

$$\begin{aligned} I &= \int \frac{\sin^3 x}{\cos^5 x} \, dx \\ &= \int \frac{\sin^2 x}{\cos^3 x} \cdot \frac{dx}{\cos^2 x} \\ &= \int u^3 \cdot du \quad \checkmark \\ &= \frac{1}{4} u^4 + C \\ &= \frac{1}{4} \tan^4 x + C \quad \checkmark \end{aligned}$$